

Generator Stationarity and Dynamical Defect-Field Geometry

Refining Structural Selection for Evolving Systems (MAAT v1.5)

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Abstract

Structural diagnostics for evolving systems must be tested against the invariants and relaxation directions of the dynamics that realise them. A support coordinate is therefore not only a static geometric feature: it is meaningful relative to a generator $\mathcal{G}$ such as a Hamiltonian, Lindbladian, field equation, solver transition rule, graph update, renormalisation map, or nonlinear flow.

This note formulates MAAT v1.5 as a generator-stationary refinement of multi-scale defect-field structural selection. The central rule is that support coordinates should measure what the dynamics preserve, damp, transport, or destabilise. In particular, balance coordinates should be stationarity-aware: they should measure compatibility with the generator rather than superficial static equality.

The immediate motivation is the open-quantum pointer benchmark of Paper 58, where stationarity balance outperforms raw population balance in predicting pointer robustness. The broader conclusion is methodological. Static supports may be useful descriptors, but structural-selection supports should survive a generator-stationarity sanity check before being treated as operational coordinates.

1 Purpose

The purpose of this note is to make structural selection less static. Many systems look balanced under a snapshot descriptor while being immediately destroyed by their own dynamics. Conversely, many robust states look asymmetric or imbalanced in static coordinates but are stationary under the generator that actually governs the system.

This is the central v1.5 update:

A structural support is meaningful only relative to the generator that defines which perturbations are preserved, relaxed, transported, or amplified.

The version label is used for continuity with the MAAT series, but the idea is general: support definitions should be validated against the dynamics of the domain. The surrounding development can be summarised as follows:

Version	Main role
v1.2.1	Four primary support sectors (H, B, S, V) with emergent robustness closure.
v1.4	Multi-scale defect-field functional beyond global scalar compression.
v1.4.1	Approximate mode-separation and explicit projection language.
v1.5	Generator-stationary structural coordinates and dynamical support validation.

Thus v1.5 keeps the v1.4 defect-field framework, but adds a stationarity test to prevent static support definitions from being imported into domains where they do not match the dynamics.

2 From Static Supports to Generator Coordinates

Let $X \in \Omega_{\text{full}}$ denote a configuration, state, graph, trajectory, or candidate solution. Let

$$\mathcal{G} : \Omega_{\text{full}} \rightarrow T\Omega_{\text{full}}$$

be the domain generator. Examples include:

- $\mathcal{G} = -i[H, \cdot]$ for closed quantum evolution,
- $\mathcal{G} = \mathcal{L}$ for Lindblad open-system evolution,
- $\mathcal{G} = \delta S / \delta \Phi$ for field-equation residuals,
- $\mathcal{G} = T_{\text{solver}}$ for a SAT solver transition rule,
- $\mathcal{G} = \Pi_{\ell \rightarrow \ell'}$ for coarse-graining or scale flow,
- $\mathcal{G} = \partial_t + \mathcal{N}$ for nonlinear dynamical systems.

A generator-aware support coordinate is not merely a scalar feature $Q_a[X]$. It is a generator-aware coordinate

$$Q_a[X; \mathcal{G}],$$

whose defect field measures local incompatibility with the relevant dynamics:

$$\boxed{d_a(x; \mathcal{G}) = \mathcal{D}_a(\mathcal{N}_x[X], \mathcal{G}X, \partial\mathcal{N}_x[X])}. \quad (1)$$

Here $\mathcal{N}_x[X]$ denotes the local neighbourhood of x , and $\partial\mathcal{N}_x[X]$ denotes its boundary or coupling interface. The support remains

$$\Gamma_a(x; \mathcal{G}) = \frac{1}{1 + d_a(x; \mathcal{G})}. \quad (2)$$

3 The Stationarity Defect

For each structural coordinate $Q_a[X; \mathcal{G}]$, define a stationarity defect:

$$\boxed{d_a^{\text{stat}}(x; \mathcal{G}) = \frac{\|\mathcal{G}Q_a(x)\|}{\Xi_a + \|Q_a(x)\|}}. \quad (3)$$

The scale $\Xi_a > 0$ prevents singular normalisation. The corresponding stationarity support is

$$\boxed{\Gamma_a^{\text{stat}}(x; \mathcal{G}) = \frac{1}{1 + d_a^{\text{stat}}(x; \mathcal{G})}}. \quad (4)$$

This definition captures the key update: a support is not high because it looks balanced at one instant. It is high when the generator does not immediately destroy the coordinate.

4 Balance as Stationarity

The most important application is the balance sector B . Earlier support definitions sometimes used static symmetry or equality. For example, in a one-qubit state with basis probabilities p_0, p_1 , a population-balance coordinate can be written as

$$B_{\text{pop}} = 1 - |p_0 - p_1|. \quad (5)$$

This is not wrong as a geometric descriptor. However, it is not generally the right balance coordinate for pointer-state robustness. Pointer-like states may be population-imbalanced while remaining dynamically stable under the environment.

The balance coordinate must be tested against the generator. For an open quantum system with density matrix ρ and Lindblad generator $\mathcal{L}$, a stationarity-sensitive balance support can be written as

$$B_{\text{stat}} = \frac{1}{1 + c_p |\dot{p}|_0 + c_\gamma \left| \frac{d}{dt} \text{Tr}(\rho^2) \right|_0}, \quad (6)$$

where $\dot{p}$ and $\frac{d}{dt} \text{Tr}(\rho^2)$ are evaluated under $\dot{\rho} = \mathcal{L}[\rho]$.

The general design rule is:

$$\boxed{B_{\text{domain}} \text{ should measure generator-compatible balance, not mere static equality.}} \quad (7)$$

5 Generator-Aware Master Functional

The v1.4 functional decomposed structural cost into mean, tail, cluster, and scale components. The generator-aware functional adds a stationarity component:

$$\boxed{F_{\text{MAAT}}^{(1.5)}[X; \mathcal{G}] = F_{\text{mean}} + F_{\text{tail}} + F_{\text{cluster}} + F_{\text{scale}} + F_{\text{stat}}.} \quad (8)$$

A representative form is

$$F_{\text{mean}} = \sum_a \lambda_a^{\text{mean}} \int d\mu(x) d_a(x; \mathcal{G}), \quad (9)$$

$$F_{\text{tail}} = \sum_a \lambda_a^{\text{tail}} q_{95}(d_a(\cdot; \mathcal{G})), \quad (10)$$

$$F_{\text{cluster}} = \sum_a \lambda_a^{\text{cl}} \frac{C_{\text{max}}^{(a)}}{N}, \quad (11)$$

$$F_{\text{scale}} = \sum_{a, \ell} \lambda_{a, \ell}^{\text{scale}} \int d\mu_\ell(x) d_a^{(\ell)}(x; \mathcal{G}), \quad (12)$$

$$F_{\text{stat}} = \sum_a \lambda_a^{\text{stat}} \int d\mu(x) d_a^{\text{stat}}(x; \mathcal{G}). \quad (13)$$

The old scalar form remains a special low-frequency projection:

$$F_{\text{MAAT}}^{\text{scalar}} = - \sum_a \lambda_a \log(\varepsilon + \bar{\Gamma}_a), \quad \bar{\Gamma}_a = \left\langle \frac{1}{1 + d_a(x; \mathcal{G})} \right\rangle_x. \quad (14)$$

Thus the generator-aware extension does not discard scalar MAAT. It identifies scalar MAAT as a compressed macro-coordinate of a richer dynamical defect-field functional.

6 Emergent Robustness

The v1.2.1 robustness hierarchy remains the default closure:

$$R_{\text{resp}} = (HBV)^{1/3}, \quad R_{\text{rob}} = \min(R_{\text{resp}}, (HBSV)^{1/4}). \quad (15)$$

The refinement changes the operational meaning of the inputs. Each sector should now be interpreted as a generator-aware support:

$$H = H[X; \mathcal{G}], \quad B = B[X; \mathcal{G}], \quad S = S[X; \mathcal{G}], \quad V = V[X; \mathcal{G}].$$

In particular, B should be stationarity-sensitive whenever the domain has a clear relaxation or evolution operator.

This preserves the v1.2.1 insight that R is emergent, not primitive. It also preserves the v1.4 insight that scalar robustness can fail in local-frustration domains such as SAT. The resulting statement is that robustness should be computed from supports whose definitions have passed a generator-stationarity sanity check.

7 Examples

Domain	Generator	Operational reading
Open quantum systems	Lindblad generator $\mathcal{L}$	A pointer-like state is balanced when $\mathcal{L}$ does not rapidly change its outcome probabilities or purity. Equal populations are not required.
Classical fields	Euler–Lagrange residual or flow operator	A field configuration has high harmony only if the residual remains small under the relevant flow; balance should include conserved-current compatibility.
SAT / CDCL	Solver transition, propagation, and conflict graph	A formula’s hardness is tied to propagation, conflict localisation, tails, and scale structure, not only to a global scalar robustness score.
Cosmology	FLRW / perturbation evolution	A viable structural observable should remain coherent under background and perturbative evolution, not merely under a present-day density snapshot.
Machine learning	Training dynamics or classifier decision map	Sample hardness should be compared to boundary motion, local label instability, and classifier response rather than feature geometry alone.

These examples illustrate the same pattern. The relevant question is not simply whether a coordinate looks orderly, but whether the domain generator treats that coordinate as persistent. For a Lindblad system this means stationarity under $\mathcal{L}$. For a SAT solver it means persistence under propagation and conflict learning. For a cosmological model it means consistency under background and perturbative evolution.

8 Operational Protocol

A generator-aware structural benchmark should follow this protocol:

1. Define the domain generator $\mathcal{G}$ before computing supports.
2. Define local defect fields $d_a(x; \mathcal{G})$ rather than only global scores.
3. Define at least one stationarity defect d_a^{stat} .
4. Compare static and stationarity-aware supports.
5. Include shuffled-defect or shuffled-sector nulls.
6. Test whether scalar compression loses tail, cluster, or scale signal.
7. Report negative transfer or failed support definitions explicitly.

This protocol is meant to protect the framework against post-hoc support construction. If a support does not survive generator-stationarity validation, it should be treated as a domain descriptor rather than as a structural selection coordinate.

9 Relation to Paper 58

Paper 58 provides the motivating benchmark. In a one-qubit Lindblad simulation, scalar stationarity balance outperformed raw population balance in predicting pointer robustness:

$$R_{\text{stat}}^2 = 0.6457, \quad R_{\text{pop}}^2 = 0.4561, \quad \Delta R^2 = +0.1896.$$

The full stationarity-sensitive defect-field model also performed best under leave-channel-family-out transfer.

The important lesson is not that MAAT solves quantum measurement. It does not. The lesson is that a naive static support can fail even when it sounds structurally plausible. In open quantum systems, the environment selects robust states through the generator. Therefore the balance coordinate must be defined relative to that generator.

10 Limitations

- This is a formal design refinement, not a microscopic derivation of the defect sectors.
- The stationarity defect depends on the chosen generator $\mathcal{G}$; in some domains the generator may be ambiguous or only approximately known.
- The decomposition $F_{\text{mean}} + F_{\text{tail}} + F_{\text{cluster}} + F_{\text{scale}} + F_{\text{stat}}$ is operational, not unique.
- Weight closure for the enlarged functional remains an open problem.
- Generator-stationarity validation does not by itself prove universality. It defines a stricter test for proposed universal supports.

11 Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

The benchmark motivating the stationarity rule is stored in:

experiments/open_quantum_pointer_paper58/

Run:

```
cd experiments/open_quantum_pointer_paper58
python3 open_quantum_pointer_paper58.py
```

This note itself is a formal framework paper. It introduces no new dataset and redistributes no external data.

12 Conclusion

MAAT v1.5 turns the defect-field programme from a static structural language into a generator-aware structural language. The key update is simple:

A support coordinate should measure what the dynamics preserve, not merely what looks balanced in a static snapshot.

This refinement keeps the v1.2.1 robustness closure and the v1.4 multi-scale defect-field functional, but adds a stationarity criterion for operational support definitions. In this sense, v1.5 is a tightening step. It reduces the freedom to invent supports after the fact and makes structural selection more falsifiable.

References

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